

HelixVPN MVP — Test Coverage Ledger

HelixVPN MVP — Test Coverage Ledger

Revision: 2

Last modified: 2026-07-06T10:45:05Z

Status: active — Phase 4 QA architecture deliverable

Authority: Constitution §11.4.25/.52/.153 (coverage ledger), §11.4.169 (mandatory test types), §11.4.6 (no guessing — gaps surfaced), §11.4.118 (enumerated coverage claim)

Scope: Every MVP requirement (HVPN-FR-NNN / HVPN-NFR-NNN) mapped to test type, owner subsystem, current evidence state, and linked Challenge / HelixQA bank IDs. Phase-2 items are tracked as P2 or NOT_APPLICABLE with an explicit re-arm trigger.

Rev 2 (2026-07-06): Pinned quantitative DDoS design targets for NFR-413/NFR-414 in v08-testing/ddos.md §10 and updated the GAP-6 closure statement to reference them. Marked GAP-3 doc-level CLOSED (measurement pending Phase-2 CHAOS region-failover drill).

How to read this ledger

Column	Meaning
Requirement	HVPN-FR-NNN / HVPN-NFR-NNN from the authoritative FR/NFR docs.
Subsystem	The HelixVPN subsystem that owns the implementation and therefore the primary test debt.
Test types	The §11.4.169 closed-set test types applied to this requirement. Legend: UNIT, INT, E2E, FA, SEC, DDOS, STRESS, CHAOS, CONC, RACE, MEM, BENCH, PERF, SCALE, UI, REC, CHAL, HQA.
Status	defined (test authored in spec/bank), automated (harness exists and runs), manual (operator-

Column	Meaning
	attended only), not-defined (no spec-level test yet), deferred-P2 (Phase-2 scoped), not-applicable-MVP (honest NA with reason).
Challenge IDs	IDs in submodules/challenges/helix_vpn/ that score this requirement.
HelixQA bank IDs	IDs in submodules/helix_qa/banks/helix_vpn/ that score this requirement.
DoD / Gate	Which MVP Definition-of-Done criterion or Phase-0 gate this requirement feeds.

Evidence-state honesty. HelixVPN is in specification / early scaffolding. No captured PASS is asserted here; every status is a *planned* state. A cell moves to automated only when a harness exists that produces a captured artifact on a clean deploy.

Legend — §11.4.169 test-type mapping

Abbrev	Type	HelixVPN evidence shape
U	unit	Pure logic: policy compiler, IPAM, delta diff, MASQUE framing, status FSM. Mocks allowed only here.
INT	integration	Real Postgres + Redis + edge booted via containers/pkg/boot; no mocks.
E2E	end-to-end	Netns + nftables + netem rig; real tunnel carries real packets.
FA	full-automation	Self-driving, -count=3, zero human-in-loop.
SEC	security	RLS, key-never-leaves, mTLS, kill-switch, DNS-leak, DPI evasion, secret audit.
DDOS	DDoS / load-flood	Handshake/volumetric flood; legit client stays usable; edge fail-static.
STR	stress	Sustained load ≥ 100 iterations / ≥ 30 s / ≥ 10 parallel.
CHAOS	chaos	SIGKILL, Redis drop, iface flap, partial write; recovery + no leak.
CONC	concurrency	Concurrent IPAM / enrollment / policy edits; no lost update.
RACE	race / deadlock	-race, loom, tsan on hot paths.
MEM	memory	iOS NE RSS, 24 h leak soak.
BENCH	benchmarking	

Abbrev	Type	HelixVPN evidence shape
		Throughput, CPU-per-Gbps, handshakes/sec, framing latency.
PERF	performance	Latency p50/p99 vs SLO budget.
SCALE	scaling	N simulated agents / 24 h soak.
UI	UI / UX	Flutter widget + golden tests.
REC	recorded-evidence	Window-scoped MP4 + vision_engine OCR verdict.
CHAL	Challenge	challenges submodule bank entry; evidence re-read.
HQA	HelixQA	helix_qa autonomous session bank entry; evidence re-read.

Functional-requirement coverage (MVP only)

A. Connect & transport (FR-0xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD Gate
HVPN-FR-001 WG crypto core	helix-core / helix-edge	U, INT, SEC, CHAL, HQA	defined	HVPN-CHAL- AUTH-TUNNEL	HVPN-HQA- AUTH-TUNNEL	G1, G2
HVPN-FR-002 plain-UDP throughput	helix-transport / edge	E2E, BENCH, PERF, FA, CHAL, HQA	not-defined	not-defined	not-defined	DoD G1
HVPN-FR-003 Transport trait	helix-core / helix-transport	U, INT, FA	not-defined	not-defined	not-defined	—
HVPN-FR-004 MASQUE/ QUIC	helix-transport / edge	E2E, SEC, BENCH, CHAL, HQA	not-defined	not-defined	not-defined	DoD G2
HVPN-FR-005 LWO	helix-transport	E2E, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-006 auto-ladder	helix-core / orchestrator	E2E, FA,	not-defined	not-defined	not-defined	DoD

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD Gate
		CHAL, HQA				
HVPN-FR-007 pin transport	helix-core / FFI	U, INT, UI	not-defined	not-defined	not-defined	—
HVPN-FR-008 port evasion	edge	E2E, INT, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-013 full-tunnel exit	helix-core / edge	E2E, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-014 split tunneling	helix-core / shims	E2E, SEC, UI	not-defined	not-defined	not-defined	—
HVPN-FR-015 transient-drop recovery	orchestrator / client	CHAOS, E2E, CHAL, HQA	defined	HVPN-CHAL-RECONNECT-ROAMING	HVPN-HQA-RECONNECT-ROAMING	—
HVPN-FR-016 MTU management	helix-transport / WG	INT, E2E	not-defined	not-defined	not-defined	—
HVPN-FR-018 status stream	helix-core / FFI / UI	U, UI, REC	not-defined	not-defined	not-defined	—
HVPN-FR-019 asymmetric per-leg default	coordinator / edge	INT, E2E	not-defined	not-defined	not-defined	—

B. Identity & enrollment (FR-1xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-101 OIDC identity	svc-identity	INT, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-102 anonymous device token	svc-identity / enrollment	INT, SEC, CHAL, HQA	not-defined	not-defined	not-defined	DoD-2
HVPN-FR-103 key-never-leaves	svc-pki / helix-core	SEC, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-FR-104 cert+token-gated enroll	svc-identity / svc-pki	SEC, INT	not-defined	not-defined	not-defined	DoD-2

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-105 short-lived mTLS cert	svc-pki	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-FR-106 control/data key separation	svc-pki / helix-core	SEC	not-defined	not-defined	not-defined	—
HVPN-FR-107 revoke < 1 s	svc-registry / edge	PERF, SEC, CHAL, HQA	not-defined	not-defined	not-defined	DoD-6
HVPN-FR-108 per-tenant CA	svc-pki	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-FR-109 cert rotation no-drop	svc-pki / helix-core	INT	not-defined	not-defined	not-defined	—
HVPN-FR-110 RLS tenant isolation	svc-identity / Postgres	SEC, INT, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-FR-111 all-roles enroll	svc-registry	INT, E2E	not-defined	not-defined	not-defined	DoD-2

C. Policy & authorization (FR-2xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-201 default-deny / fail-closed	svc-policy / edge	SEC, E2E, META, CHAL, HQA	not-defined	not-defined	not-defined	DoD-3
HVPN-FR-202 ACL compiler grammar	svc-policy	U, INT	not-defined	not-defined	not-defined	—
HVPN-FR-203 AllowedIPs + verdict map	svc-policy / edge	E2E, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-204 authorized	edge / svc-policy	E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-3

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
reach + denied						
HVPN-FR-205 policy edit < 1 s	svc-policy / coordinator	PERF, E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-5
HVPN-FR-206 split-horizon default	svc-policy / IPAM	E2E, SEC	not-defined	not-defined	not-defined	—
HVPN-FR-207 need-to-know map filtering	svc-coordinator	SEC, META	not-defined	not-defined	not-defined	—

D. Routing, addressing & multi-network (FR-3xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-301 1→N overlay	IPAM / routing / edge	E2E, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-FR-302 overlapping-CIDR resolve	svc-ipam / routing	E2E	not-defined	not-defined	not-defined	—
HVPN-FR-303 ULA /48 + 4via6	svc-ipam / routing	E2E, INT	not-defined	not-defined	not-defined	—
HVPN-FR-304 stable overlay IP	svc-ipam	INT	not-defined	not-defined	not-defined	—
HVPN-FR-305 connector advertise + route	svc-registry / edge	E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-3
HVPN-FR-306 two-way no-inbound	edge / helix-core	E2E, SEC	not-defined	not-defined	not-defined	DoD-3
HVPN-FR-307 IPAM concurrent alloc	svc-ipam	STR, CONC, CHAL, HQA	not-defined	not-defined	not-defined	—

E. Kill-switch & leak protection (FR-5xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-501 core-owned kill-switch	helix-core / shims	U, SEC, CHAL, HQA	not-defined	not-defined	not-defined	DoD-7
HVPN-FR-502 no plaintext on drop	helix-core / OS firewall	SEC, E2E, CHAL, HQA	defined	HVPN-CHAL-KILL-SWITCH	HVPN-HQA-KILL-SWITCH	DoD-7
HVPN-FR-503 DNS through tunnel	helix-core / resolver	SEC, E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-7
HVPN-FR-504 per-OS firewall	platform shims	SEC, per-OS branch	not-defined	not-defined	not-defined	DoD-7

F. Console & administration (FR-6xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-601 CRUD + RBAC	svc-api / Console	INT, E2E, SEC, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-FR-602 Console no-core build	helix-ui / build	INT, FA	not-defined	not-defined	not-defined	—
HVPN-FR-603 live WS/SSE	svc-api	E2E	not-defined	not-defined	not-defined	—
HVPN-FR-604 live topology view	Console	UI, E2E	not-defined	not-defined	not-defined	—
HVPN-FR-605 control-action audit	svc-telemetry / audit	SEC, UI	not-defined	not-defined	not-defined	—
HVPN-FR-606 multi-tenant isolation	Console / RLS	SEC, CHAL, HQA	not-defined	not-defined	not-defined	—
						—

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-608 responsive + visual	Console / helix-ui	UI, REC	not-defined	not-defined	not-defined	
HVPN-FR-609 exportable audit slice	audit / svc-telemetry	SEC, INT	not-defined	not-defined	not-defined	—

G. Connector (FR-7xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-701 outbound-only connector	helix-core (advertise/route)	E2E, SEC, CHAL, HQA	not-defined	not-defined	not-defined	DoD-2
HVPN-FR-702 headless daemon	helix-core / connector shim	E2E, INT	not-defined	not-defined	not-defined	—
HVPN-FR-703 shared Rust core	helix-core	INT, FA	not-defined	not-defined	not-defined	—
HVPN-FR-704 advertise + route	svc-registry / edge	E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-3
HVPN-FR-707 availability-following on drop	orchestrator / connector	CHAOS, E2E, CHAL, HQA	not-defined	not-defined	not-defined	—

H. Observability & telemetry (FR-8xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-801 no durable flow table	schema / migrations	SEC, META, CHAL, HQA	not-defined	not-defined	not-defined	DoD-8
HVPN-FR-802 ephemeral presence	svc-events / Redis	INT	not-defined	not-defined	not-defined	—
		INT				—

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-803 Prometheus metrics	svc-telemetry		not-defined	not-defined	not-defined	
HVPN-FR-804 convergence SLO metrics	svc-telemetry	PERF, INT	not-defined	not-defined	not-defined	—
HVPN-FR-805 counts-only success metrics	svc-telemetry	SEC, INT	not-defined	not-defined	not-defined	—

I. Deployment & self-host (FR-9xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-901 helixvpncctl init	helixvpncctl / deploy	E2E, FA, CHAL, HQA	not-defined	not-defined	not-defined	DoD-1
HVPN-FR-902 rootless Podman	deploy / quadlets	SEC, E2E	not-defined	not-defined	not-defined	—
HVPN-FR-904 helixvpncctl commands	helixvpncctl	E2E, INT	not-defined	not-defined	not-defined	—
HVPN-FR-906 hardened edge container	deploy / edge	SEC, E2E	not-defined	not-defined	not-defined	—
HVPN-FR-907 fail-static on CP down	edge / coordinator	CHAOS, E2E, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-FR-908 build-essential regeneration	repo tooling	U, FA	not-defined	not-defined	not-defined	—

J. Clients & platform apps (FR-10xx)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-FR-1001 three flavors one tree	helix-ui / Melos	INT, FA, UI	not- defined	not- defined	not- defined	—
HVPN-FR-1002 shared core + FFI stream	helix-core / FFI	INT, UI, REC	not- defined	not- defined	not- defined	—
HVPN-FR-1003 one- button connect	helix-ui	UI, REC, E2E, CHAL, HQA	defined	HVPN-CHAL - CLIENT-UI - VISUAL	HVPN - HQA - CLIENT - UI - VISUAL	DoD-8
HVPN-FR-1004 iOS NE memory	shim-apple / helix-core	MEM, CHAL, HQA	not- defined	not- defined	not- defined	G3
HVPN-FR-1005 Android VpnService	shim-android	E2E, UI, REC	not- defined	not- defined	not- defined	—
HVPN-FR-1006 Linux kernel WG	shim-linux	E2E, UI	not- defined	not- defined	not- defined	—
HVPN-FR-1012 exit/ network + obfuscation UI	helix-ui / FFI	UI, E2E	not- defined	not- defined	not- defined	—
HVPN-FR-1013 light + dark themes	helix-ui / OpenDesign	UI, REC	not- defined	not- defined	not- defined	—
HVPN-FR-1014 three apps drive E2E	all clients	E2E, UI, REC, CHAL, HQA	not- defined	not- defined	not- defined	DoD-8

Non-functional-requirement coverage (MVP only)

Performance (NFR-001...009)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD Gate
HVPN-NFR-001 ≥ 80 % bare-link throughput	helix-transport / edge	PERF, BENCH, E2E, CHAL, HQA	not-defined	not-defined	not-defined	G1
HVPN-NFR-002 ≥ 50 % MASQUE throughput	helix-transport / edge	PERF, BENCH, E2E	not-defined	not-defined	not-defined	G2
HVPN-NFR-003 p99 convergence < 1 s	svc-coordinator	PERF, INT, CHAL, HQA	not-defined	not-defined	not-defined	DoD-
HVPN-NFR-004 revoke < 1 s	svc-registry / edge	PERF, SEC	not-defined	not-defined	not-defined	DoD-
HVPN-NFR-005 WG handshake latency	helix-core / edge	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-006 ladder escalation bounded	orchestrator / ladder	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-008 MTU 1420 / no fragment	helix-transport	INT, E2E	not-defined	not-defined	not-defined	—
HVPN-NFR-009 admin-perceived edit < 1 s	svc-api $\rightarrow$ coordinator	E2E, FA, PERF	not-defined	not-defined	not-defined	DoD-

Scale (NFR-100...108)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-100 ≥ 10 k streams	svc-coordinator	SCALE, STR, MEM	not-defined	not-defined	not-defined	SLO4
HVPN-NFR-101 bounded coordinator memory	svc-coordinator	SCALE, MEM	not-defined	not-defined	not-defined	SLO4
HVPN-NFR-102 minimal affected set	svc-coordinator	U, PERF	not-defined	not-defined	not-defined	—
HVPN-NFR-103 connectors/tenant	svc-ipam / edge	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-104 1 user $\rightarrow$ N nets	routing / edge	E2E	not-defined	not-defined	not-defined	—
HVPN-NFR-105 reconnect storm	svc-coordinator / edge	STR	not-defined	not-defined	not-defined	—
HVPN-NFR-106 per-tenant isolation	svc-coordinator	CONC, RACE	not-defined	not-defined	not-defined	—

Availability & HA (NFR-200...207)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-200 fail-static CP down	edge	CHAOS, E2E, CHAL, HQA	defined	HVPN-CHAL-CONTROL-PLANE-HA	HVPN-HQA-CONTROL-PLANE-HA	—
HVPN-NFR-201 Postgres/Redis blip	svc-coordinator	CHAOS	not-defined	not-defined	not-defined	—
		CHAOS				—

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-202 Redis loss graceful	svc-coordinator / presence		not-defined	not-defined	not-defined	
HVPN-NFR-203 no-work-loss consumer	svc-events	CHAOS, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-204 coordinator restart resume	svc-coordinator	CHAOS, INT, CHAL, HQA	not-defined	not-defined	not-defined	—
HVPN-NFR-207 poison → DLQ	svc-events	CHAOS, INT	not-defined	not-defined	not-defined	—

Privacy (NFR-300...309)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-300 no durable flow table	schema / migrations	SEC, META	not-defined	not-defined	not-defined	DoD-8
HVPN-NFR-301 schema-lint not tautology	schema lint	META	not-defined	not-defined	not-defined	DoD-8
HVPN-NFR-302 bus carries no traffic shape	svc-events	SEC, payload-lint	not-defined	not-defined	not-defined	—
HVPN-NFR-303 audit = control-only	svc-telemetry	U, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-304 ephemeral presence	svc-events / Redis	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-305	svc-telemetry	U, INT	not-defined	not-defined	not-defined	—

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
coarse last_seen_at						
HVPN-NFR-306 counts-only metrics	svc-telemetry	U	not-defined	not-defined	not-defined	—
HVPN-NFR-307 anonymous identity	svc-identity	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-308 runtime no-log signature	migrations / deploy	E2E, CHAL, HQA	not-defined	not-defined	not-defined	DoD-8
HVPN-NFR-309 audit retention policy	audit / svc-telemetry	INT	not-defined	not-defined	not-defined	—

Security (NFR-400...414) — including GAP-6 closure

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-400 WG crypto never forked	helix-core / CI	SEC, META	not-defined	not-defined	not-defined	—
HVPN-NFR-401 outbound-only edges	edge / deploy	SEC	not-defined	not-defined	not-defined	—
HVPN-NFR-402 need-to-know map	svc-coordinator	E2E, META	not-defined	not-defined	not-defined	—
HVPN-NFR-403 mTLS + revoke	svc-pki / edge	SEC	not-defined	not-defined	not-defined	—
	helix-core / shims	SEC, E2E	not-defined	not-defined	not-defined	DoD-7

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-404 kill-switch no leak						
HVPN-NFR-405 DNS-leak protection	helix-core / resolver	SEC, E2E	defined	HVPN-CHAL-DNS-LEAK	HVPN-HQA-DNS-LEAK	DoD-7
HVPN-NFR-406 private key on-device	helix-core / FFI	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-408 RLS isolation	Postgres / svc-identity	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-409 anti-bluff gauntlet	QA harness	META	not-defined	not-defined	not-defined	—
HVPN-NFR-410 DPI evasion	helix-transport	E2E	not-defined	not-defined	not-defined	DoD-4
HVPN-NFR-411 secrets rotation	deploy / KMS	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-412 key rotation cadence	svc-pki / helix-core	SEC, INT	not-defined	not-defined	not-defined	—
HVPN-NFR-413 control-plane API rate limiting	svc-api / Redis	DDOS, STR, SEC	defined	HVPN-CHAL-NFR413-API-Rate-Limit	HVPN-HQA-NFR413-API-Rate-Limit	GAP-6 owner
HVPN-NFR-414 data-plane DDoS resilience	edge / helix-transport	DDOS, SEC	defined	HVPN-CHAL-NFR414-Edge-Flood	HVPN-HQA-NFR414-Edge-Flood	GAP-6 owner

GAP-6 closed. The DDOS test type now has explicit ownership: **NFR-413** (control-plane API gateway, token-bucket rate limiting per API key / device) and **NFR-414** (data-plane edge, UDP/handshake-flood resilience and fail-static behavior). Both are defined with parked Phase-2

banks; the MVP cell is not-applicable-MVP (single-node-selfhost) with a mechanical re-arm trigger when topology becomes multi-tenant-ha.

Resource (NFR-500...506)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-500 iOS NE memory headroom	shim-apple / helix-core	MEM, CHAL, HQA	not-defined	not-defined	not-defined	G3
HVPN-NFR-501 battery-sensitive	helix-core / platform shims	BENCH	not-defined	not-defined	not-defined	—
HVPN-NFR-502 binary footprint	helix-core	BENCH	not-defined	not-defined	not-defined	—
HVPN-NFR-503 ≤ 60 % host RAM	build tooling	BENCH	not-defined	not-defined	not-defined	—
HVPN-NFR-504 bounded CP memory	svc-coordinator	SCALE, MEM	not-defined	not-defined	not-defined	SLO4
HVPN-NFR-505 lightweight sampler	svc-telemetry	BENCH	not-defined	not-defined	not-defined	—
HVPN-NFR-506 bounded send queues	svc-coordinator	STR, MEM	not-defined	not-defined	not-defined	—

Portability (NFR-600...609)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-600 three flavors one tree	helix-ui / Melos	INT, FA, UI	not-defined	not-defined	not-defined	—

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-601 shared Rust core	helix-core	INT, FA	not-defined	not-defined	not-defined	—
HVPN-NFR-602 iOS/Android/Linux Access	platform shims	E2E, FA	not-defined	not-defined	not-defined	DoD-8
HVPN-NFR-603 Connector headless	helix-core / shims	INT	not-defined	not-defined	not-defined	—
HVPN-NFR-604 Console Web + desktop	helix-ui / Console	E2E, UI	not-defined	not-defined	not-defined	—
HVPN-NFR-609 cross-platform parity	platform shims	per-OS branch	not-defined	not-defined	not-defined	—

Operability & Compatibility (NFR-700...703)

Requirement	Subsystem	Test types	Status	Challenge IDs	HelixQA bank IDs	DoD / Gate
HVPN-NFR-702 protobuf / OpenAPI compatibility	codegen pipeline	U, META	not-defined	not-defined	not-defined	—

MVP Definition-of-Done acceptance-criteria coverage

DoD	Criterion	Satisfying FRs/NFRs	Primary challenge	Primary HelixQA bank
DoD-1	Self-host from zero	FR-901, FR-902	not-defined	not-defined

DoD	Criterion	Satisfying FRs/NFRs	Primary challenge	Primary HelixQA bank
DoD-2	Enroll Connector + Client	FR-102, FR-104, FR-111, FR-701	HVPN-CHAL - AUTH-TUNNEL	HVPN-HQA - AUTH-TUNNEL
DoD-3	Reach authorized / deny unauthorized	FR-201, FR-204, FR-301, FR-305, FR-306, FR-704	not-defined	not-defined
DoD-4	Auto-escalate to MASQUE	FR-004, FR-006, NFR-410	not-defined	not-defined
DoD-5	Policy edit < 1 s	FR-205, NFR-003, NFR-009	not-defined	not-defined
DoD-6	Revoke < 1 s	FR-107, NFR-004	not-defined	not-defined
DoD-7	Kill-switch + DNS-leak	FR-501...504, NFR-404, NFR-405	HVPN-CHAL - KILL-SWITCH	HVPN-HQA - KILL-SWITCH
DoD-8	No durable log + three apps	FR-801, FR-1001, FR-1014, NFR-300, NFR-308	HVPN-CHAL - CLIENT-UI-VISUAL	HVPN-HQA - CLIENT-UI-VISUAL

Phase-0 gate coverage

Gate	Question	Satisfying FRs/NFRs	Primary challenge	Primary HelixQA bank
G1	Plain-UDP WG throughput	FR-002, NFR-001	not-defined	not-defined
G2	MASQUE/QUIC through DPI block	FR-004, NFR-002	not-defined	not-defined
G3	iOS NE memory ceiling	FR-1004, NFR-500	not-defined	not-defined
G4	Go-vs-Rust edge benchmark	FR-003, NFR-001/002	not-defined	not-defined
G5				

Gate	Question	Satisfying FRs/NFRs	Primary challenge	Primary HelixQA bank
G6	flutter_rust_bridge FFI	FR-018, FR-1002	not-defined	not-defined
	Push-based reconcile	FR-205, NFR-003	not-defined	not-defined

GAP-6 closure statement

Gap: requirements-traceability.md Rev 4 identified that the DDOS §11.4.169 test type was defined in the legend but traced to zero requirements.

Closure in this ledger:

1. **Requirement HVPN-NFR-413** (control-plane API rate limiting) is explicitly owned by **svc-api** and verified by **DDOS** + stress + security tests.
2. **Requirement HVPN-NFR-414** (data-plane DDoS/UDP-flood/amplification resilience) is explicitly owned by **helix-edge** / **helix-transport** and verified by **DDOS** + security tests.
3. Both rows list concrete, minted Challenge IDs (HVPN-CHAL-NFR413-API-Rate-Limit, HVPN-CHAL-NFR414-Edge-Flood) and HelixQA bank IDs (HVPN-HQA-NFR413-API-Rate-Limit, HVPN-HQA-NFR414-Edge-Flood) that resolve in the committed banks.
4. The bank entries are **defined**; the live DDoS attack harness is not exercised in the single-node-selfhost MVP topology and mechanically re-arms when `deployment.topology == "multi-tenant-ha"` in Phase 2.

Quantitative design targets pinned (see v08-testing/ddos.md §10). The Phase-2 DDoS bank will calibrate against the following design targets, not measured baselines:

1. **NFR-413 control-plane API flood:** `ATTACK_PPS_API` = 10 000 invalid req/s for 10 s; legitimate-client SLO = p95 < 200 ms + zero HTTP 5xx.
2. **NFR-414 data-plane edge WG-init flood:** `ATTACK_PPS_EDGE` = 100 000 pps for 10 s; legitimate-handshake p99 budget < 2 × unflooded baseline.
3. **Edge fail-static (EDGE_FAILSTATIC):** edge stays up, or restarts within 5 s with **zero** plaintext leak.

These targets will be calibrated during the Phase-2 multi-tenant-ha soak; the DDOS gate becomes release-blocking only against the calibrated value.

Coverage summary

Metric	Count
MVP functional requirements traced	53
MVP non-functional requirements traced	38
Requirements with minted Challenge / HelixQA IDs	8
Total test-type cells mapped	400+ (planned)
Challenge IDs minted	8
HelixQA bank IDs minted	8
DoD criteria covered	8 / 8 (3 with minted primary bank IDs)
Phase-0 gates covered	6 / 6 (0 with minted primary bank IDs)
Gaps closed (this pass)	GAP-1, GAP-3 (doc-level), GAP-4, GAP-5 (doc-level), GAP-6
Remaining open gaps	None at documentation level. Quantitative targets for NFR-205 (RTO/RPO) and NFR-413/414 (DDoS) remain measurement-pending until Phase-2 drills/benchmarks run against built code.
Doc-level closed / measurement-pending	GAP-3 — DR targets/runbooks authored; GAP-5 — all v08-testing/ docs authored; evidence states honestly PENDING in spec phase

Provenance

- Sources: docs/research/mvp/final/10-testing-acceptance-and-qa.md, docs/research/mvp/final/v01-product/functional-requirements.md, docs/research/mvp/final/v01-product/nonfunctional-requirements.md, docs/research/mvp/final/v00-meta/requirements-traceability.md, docs/research/mvp/final/v08-testing/ddos.md.
- Challenge / HelixQA bank conventions derived from submodules/challenges/banks/examples/*.json, submodules/challenges/banks/yole/*.yaml, submodules/helix_qa/banks/*.yaml.

- Constitution bindings: §11.4.6, §11.4.25, §11.4.52, §11.4.118, §11.4.153, §11.4.169.